Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/20/2005
PI Name:	Dreizin, Edward L. Ph.D.		
Project Title:	High Temperature Phases and Phase Equilibria in Reactive Molten Metal-Based Systems		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Glasses and ceramics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	07102-1982	**Congressional District:**	10
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-05
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
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No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Law, C. (Princeton University) Keil, D. G. (Titan Corporation)		
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Containerless microgravity experiments will be designed and conducted in which solid components (e.g., powders) premixed at room temperature will be heated using a CO2 laser in a controlled pressure inert gas or vacuum environment, kept at a high temperature for a period of time, and rapidly quenched. The initial components will include pure metals, solid metal–gas solution phases (e.g., áZr produced by the annealing of a pure metal in an oxygen and/or nitrogen environment), carbon and metal carbides, oxides, and nitrides. The rate of the sample heating, the maximum temperature, and the time at the maximum temperature will be the parameters varied independently in the microgravity experiments. A rapid quenching technique, splat quenching, will be employed to achieve a fast cooling rate. Phases produced at the high temperatures will be analyzed using both real time observation of the sample behavior, optical temperature measurements, and, most importantly, detailed analyses of the quenched samples using x-ray diffraction and electron microscopy techniques. Quenched samples will be subjected to various annealing cycles to address the

Task Description:	stability of the detected phases at different temperatures. The general objective of this program is to experimentally characterize the high-temperature phase compositions and phase equilibria in molten metal-based binary and ternary systems. The very wide range of materials for which the high-temperature equilibria are not well known requires us to specify selected systems that will be addressed in this proposed four-year research project. We base our selection on the applications that we are most familiar with (i.e., high-energy density materials) and also on the materials science literature that we are aware of that describes materials currently being actively investigated for other applications. With this rationale we chose boron-carbon-oxygen and zirconium-oxygen-nitrogen systems as the first two candidates. The results of this research will be compiled as stable and metastable phase diagram cross-sections (at selected temperatures) for the materials explored. This information will be valuable for both fundamental understanding of the nature of these materials and also for their practical use in many areas. New high-energy density metal-based materials will be developed that will consist of metastable metal-oxygen (e.g., Zr-O, B-O, Al-O, etc.) solutions. In such materials, fuel and oxidizer will be premixed on the atomic level to provide a fast combustion rate and a significantly enhanced performance of propellants, explosives, and incendiaries. The performance enhancement will be achieved by exploiting phase transitions occurring in the liquid, high-temperature, igniting or burning metal-gas systems. Thus, reliable information on the phase equilibria and transitions occurring in such systems will lead directly to better performance control of the presently available metallized energetic materials and to the design of new, more effective compositions. Similar information on the high-temperature phase equilibria becomes vital for many other rapidly developing technological areas. New ionic conductors based on zirconium oxynitrides and doped zirconium oxides are being developed for use in oxygen sensors and fuel cells. These devices are often employed at elevated temperatures at which the stability of the known oxynitride phases is not well known. The high-temperature equilibria addressed in this proposed research will provide information needed to assess this phase stability, predict the stable and metastable phases that can be formed, critically evaluate and modify the material synthesis and processing techniques. Compositions of the precursors employed in manufacturing of zirconia based ionic conductors will be adjusted and the thermal cycling used in their production and testing technologies will also be improved using the results of this proposed program. Knowledge of the high-temperature phase equilibria for the boron-oxygen and boron-carbon-oxygen systems is essential for development and fabrication of many new semiconductors, glasses, and ultra-hard materials. Extension of the metal-gas binary and ternary phase diagrams into the high-temperature range is clearly of generic scientific interest and will improve our understanding of the fundamental mechanisms of interaction between metal and gas atoms in various solid and liquid solutions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	The microgravity experiments were performed onboard KC-135 aircraft. The results were analyzed and presented in a peer-reviewed article. The project has been completed.
Bibliography Type:	Description: (Last Updated: 10/14/2009)
Articles in Peer-reviewed Journals	Ermoline A, Schoenitz M, Hoffmann VK, Dreizin EL. "Experimental technique for studying high-temperature phases in reactive molten metal based systems." Review of Scientific Instruments 2004 Dec :5177-85. http://dx.doi.org/10.1063/1.1819011 , Dec-2004